

The change specification problem

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1 Introduction

In recent work Gorman and Reiss (2026) give a formal definition to the *feature specification problem* (FSP), the problem of determining the natural class representations of the targets and triggers of phonological processes. They argue that the received wisdom—that the FSP is solved by finding the minimal (in terms of the number of valued features) specification—is incorrect, and suggest a simple alternative, in which natural class specifications are determined via intersection. In addition to targets and triggers, traditional phonological rules are also defined by the *change*, or the modification applied to the targets of the rule. In Logical Phonology (e.g., Bale et al. 2014, Bale and Reiss 2018, Bale et al. 2020), these changes are defined by an operator and its right-hand side argument, a set of valued (binary) features. However, whereas the proper solution to the FSP is contentious—see Gorman and Reiss 2026—the *change specification problem* (CSP) is comparatively simple. I provide a solution below for LP’s theory of possible rules, but the basic observations generalize to other theories, such as the rule system of classical generative phonology (e.g., Chomsky and Halle 1968, henceforth *SPE*).

2 The algorithm

The change specification algorithm, like the feature-specification algorithm, is a subproblem arising in the acquisition of phonological grammars, and is specifically encountered while inducing individual rules. The input to the change specification is then a list of pair of segments—that is, sets of valued features—where the first element in each pair represents the segment before the putative rule applies and the second is the corresponding segment after. Let us also assume that in no pair are the “before” and “after” element identical. Let us call this data structure the *correspondence set*. If this set of correspondences defines a valid rule, then the “before” segments will necessarily comprise a natural class. However, it is not necessarily the case that the “after” segments will form a natural class.¹

The algorithm proceeds to compute bidirectional set difference on each of the correspondence

¹Thanks to Malek Azadegan (p.c.) for pointing this out to me.

pairs. That is, for each pair $x \rightsquigarrow y$, it computes $c_1 = x \setminus y$ and $c_2 = y \setminus x$.² If c_1 and c_2 do not take on consistent values for each pair in the correspondence set, then there is no valid LP rule relating the correspondents. However, if $c_1 = \emptyset$ then the change is $\sqcup c_2$, part of a unification rule. And, if $c_2 = \emptyset$, then the change is $\setminus c_1$, part of a subtraction rule.

3 Examples

First consider the following correspondence sets, where the capitalized segments represent oral stops underspecified for VOICE but otherwise identical to the corresponding lowercase symbols.

- a. $\{b \rightsquigarrow B, d \rightsquigarrow D, g \rightsquigarrow G\}$
- (1) b. $\{B \rightsquigarrow p, D \rightsquigarrow t, G \rightsquigarrow k\}$
- c. $\{b \rightsquigarrow p, d \rightsquigarrow t, g \rightsquigarrow k\}$

In (1a), $c_1 = [+VOICE]$ and $c_2 = \emptyset$, so the change is given by $\setminus [+VOICE]$. In (1b), $c_1 = \emptyset$ and $c_2 = [-VOICE]$, so the change is given by $\sqcup [-VOICE]$. In (1c), for the first pair, $c_1 = [+VOICE]$ and $c_2 = [-VOICE]$. This does not define a valid LP rule by the above construction, and it should be obvious why: this would be a *feature-changing* rule of the sort familiar from *SPE*. In LP, such rules are not *per se* statable, but feature-changing processes can be derived by ordering a subtraction rule before a unification rule. Indeed, the changes for (1a,b) above—namely, $\setminus [+VOICE]$ and $\sqcup [-VOICE]$, applied in that order—would have the desired effect.

Let’s now consider a more complex example from a real language. In Breton, there is a process of spirantization triggered by a highly-specific set of immediately-preceding determiner-like heads, including certain possessive proclitics (e.g., $[m\acute{a}]$ ‘my’) and numerals (e.g., $[te.i\text{ɤ}]$ ‘three’).

(2) Breton spirantization (Timm 1985, Iosad 2014):

- a. $pl\acute{a}:x$ ‘girl’ $te.i\text{ɤ} fl\acute{a}:x$ ‘three girls’
- b. $ta:\text{ɖ}$ ‘father’ $m\acute{a} za:\text{ɖ}$ ‘my father’
- c. $ka:z$ ‘cat’ $m\acute{a} ha:z$ ‘my cat’

This sets up the following (incomplete) correspondence set:

- (3) $\{p \rightsquigarrow f, t \rightsquigarrow z, k \rightsquigarrow h\}$

For concreteness, I assume the following partial featural specification for these segments:³

	p	f	t	z	k	h
(4) VOICE	–	–	–	+	–	–
CONTINUANT	–	+	–	+	–	+
LABIAL	+	+	–	–	–	–
CORONAL	–	–	+	+	–	–
DORSAL	–	–	–	–	+	–

²Alternatively, one could first compute the generalized intersection of all “before” and all “after” segments, respectively, and then compute the bidirectional set difference between the two intersections. However, this alternative procedure does not allow one to easily detect inconsistencies in c_1, c_2 , making it less suitable for a CSP algorithm.

³I put aside features like CONSONANTAL, LABIODENTAL, SPREAD GLOTTIS, and STRIDENT, all of which might be relevant, as these simply serve to make the solution more verbose. As is common, I assume /h/ is placeless, but nothing depends on this assumption.

We then obtain the following set differences about these featural specifications:

- a. $p \rightsquigarrow f$: $c_1 = \{-\text{CONTINUANT}\}$ $c_2 = \{+\text{CONTINUANT}\}$
- (5) b. $t \rightsquigarrow z$: $c_1 = \begin{Bmatrix} -\text{VOICE} \\ -\text{CONTINUANT} \end{Bmatrix}$ $c_2 = \begin{Bmatrix} +\text{VOICE} \\ +\text{CONTINUANT} \end{Bmatrix}$
- c. $k \rightsquigarrow h$: $c_1 = \begin{Bmatrix} -\text{CONTINUANT} \\ +\text{DORSAL} \end{Bmatrix}$ $c_2 = \{+\text{CONTINUANT}\}$

Clearly, this cannot be stated as a single LP rule. First, separate subtraction and unification rules are needed to handle the “swap” from $-\text{CONTINUANT}$ to $+\text{CONTINUANT}$, much like in (1c) above. Of course, this is not the only problem for constructing a unified change specification: there is also a VOICE “swap” in (5b) not shared with (5a,c), and there is a deletion of $+\text{DORSAL}$ in (5c) not shared with (5a,b). The following appears to be the minimal set of rules—I omit the environment, which is contentious, but believed to be at least partially morphemic and/or morphosyntactic—needed to derive the outputs:

- (6) $[\text{k}] \setminus \{+\text{DORSAL}\}$
- (7) $[\text{t}] \setminus \{-\text{VOICE}\}$
- (8) $[\] \setminus \{-\text{CONTINUANT}\}$
- (9) $[\] \sqcup \{+\text{CONTINUANT}\}$
- (10) $[\] \sqcup \{+\text{VOICE}\}$
- (11) Critical ordering: (8) $\ll$ (9), (7) $\ll$ (10)

In contrast, classical generative phonology, which has both feature-changing rules and the angled-brace notation for adding additional if-else conditions to rules, can state this as a single rule. Speaking of a similar process in Welsh, the so-called “soft mutation”, Zwicky (1974:222) writes that “the notational tricks used to achieve this solution ought to be disallowed”. LP takes a similar position, reflecting its commitment to ontological austerity.

4 Generalizations

As mentioned above, LP uses *segment rules* for insertion and deletion of whole segments. These rules are defined using the $\mapsto$ operator, a placeholder ϵ , and linking the inserted or deleted segmental feature bundle to an X slot.

- (12) Insertion schema:

$$\begin{array}{c} \text{X} \\ | \\ \epsilon \mapsto \{ \dots \} / [\dots] \text{ — } [\dots] \end{array}$$

(13) Deletion schema:

$$\begin{array}{c} X \\ | \\ [\dots] \mapsto \epsilon / [\dots] \text{ --- } [\dots] \end{array}$$

Suppose we allow the correspondent set to include ϵ , either on the left or right, in addition to segments. Furthermore, let us stipulate that $\epsilon \setminus S = \epsilon$ and $S \setminus \epsilon = S$ for any segment S . Then, a valid insertion change is present when $c_1 = \epsilon$ and c_2 is a single consistent segment. For deletion rules, all that is necessary is that $c_2 = \epsilon$, though for the rule itself to be valid, c_1 (the target) must also comprise a natural class (see Gorman and Reiss 2025) rather than a consistent set. This asymmetry arises because deletion rules delete natural classes whereas insertion rules merely insert segments.

References

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